

SPETC v10 — User Guide

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2007–2026

Abstract

This guide explains how to install, configure and operate `SPETC`, the spectro-photometric exposure time calculator, from the first launch to photometric and spectroscopic planning, results interpretation and CSV export. No knowledge of the internal physics is required; the companion document *Physics of the ETC* covers the models and formulae.

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1 Getting started

1.1 Requirements and installation

SPETC needs Python 3.10 or later with Tk support. Install the dependencies and start the program from the distribution directory:

```
python3 -m pip install -r requirements.txt
python3 etc_gui.py
```

The `data/` directory shipped with the program must stay beside `etc_gui.py`; it contains the stellar template catalogue (`interpola.db.csv` and the CALSPEC/BPGS spectra), the default detector QE curve (`qe.dat`), the SVO filter profiles with their list (`filters.list`), the optional `BLANK.xml` unfiltered passband and the optional Earth-atmosphere FITS table. The data directory can also be pointed elsewhere from the Stellar template selector panel.

1.2 First launch

On start-up SPETC restores the last saved configuration from `etc_user_config.json` (created automatically; the default preset is Asiago/Cima Ekar) and the most recently used instrument profile, if any. The configuration is re-saved after every successful calculation and on normal exit, so the program always reopens as you left it.

1.3 A one-minute first calculation

1. Pick a site from the Observatory selector (or type latitude, longitude, elevation).
2. Set the date and UT time, or leave the current time.
3. Type target coordinates (decimal or sexagesimal), or type a name and press **Resolve name** (needs internet; nothing else ever does).
4. Select an observing filter and a stellar template; enter the target magnitude.
5. Press **Run ETC**. The Results window opens with the numbers; the plot window shows S/N versus time, altitude, and the sky path.

2 The main window

The window is organised in five columns, left to right.

2.1 Observation, site and target

Site selector (built-in observatories merged with your persistent `etc_sites.json`; **Save current site** and **Import site** file manage it), geographic coordinates and elevation, date and UT time of the planned observation, the observing-interval selector, the time zone (IANA name with DST, e.g. Europe/Rome, or a fixed UTC offset), and the target: RA/Dec fields (decimal degrees or `hh:mm:ss / dd:mm:ss`) with the optional **SIMBAD Resolve name** button. Running the ETC itself never uses the network: only the coordinates in the fields count.

2.2 Telescope, detector and conditions

Telescope diameter, central obstruction, focal length and scalar optics throughput; optional calibrated instrument-response and slit resolving-power curves (two-column files with an explicit wavelength unit); detector pixel size, gain, full well, ADC bit depth, read noise and dark current; seeing FWHM; the sky-background mode (`ing` = physical night/twilight/day model, `fixed_ab` = your own observed sky surface brightness); and the **PSF model** (`gaussian` or `moffat` with its β ; Moffat reproduces the wings of real star images and is recommended). **Save/Load instrument profile** stores the whole column as a portable JSON with its QE and response curves.

2.3 Calculation settings

Photometry or spectroscopy; exposure time; optional **target S/N** (when set, the ETC returns the required exposure instead, flagging saturation). The photometry panel adds the aperture radius and the **source geometry**: `point`, or `extended` with the source area in arcsec^2 for galaxies, nebulae and planets (the magnitude you enter stays the *total* integrated magnitude). The spectroscopy panel holds the resolving power, slit width, extraction height, sampling, the slitless parameters (Sect. 4), the wavelength range and the S/N reference wavelength, and the **slit orientation** (`parallactic` or `fixed`).

2.4 Filter and template selection

Two independent filters: the **reference** filter, in which your magnitude is expressed (Vega or AB), and the **observing** filter, physically in the beam for both photometry and spectroscopy (spectroscopy defaults to BLANK, the unfiltered response). The panel previews the observing transmission curve with its pivot wavelength, width and zero point; the template selector searches the catalogue by name and previews the selected flux distribution; **Validate V** and **B–V** checks the template’s synthetic colours against the catalogue.

2.5 Run actions

Run ETC, the live observatory status (Sun/Moon altitude, current airmass of the target), and Exit (which saves the configuration).

3 Photometry step by step

3.1 Point sources

1. Set **Source geometry** to `point`.
2. Choose the aperture radius. As a rule of thumb $1.5\times$ the seeing FWHM captures $\sim 95\%$ of a Moffat profile while limiting sky noise; the ETC applies the exact encircled-energy correction, so any radius gives a correct answer.
3. Enter the target magnitude and its reference filter and system (Vega default; AB available).
4. Run. The S/N-versus-time curve shows the whole night; the red marker is your selected time.

For bright stars do not be surprised that the S/N stops growing with brightness at a few $\times 10^3$: atmospheric scintillation sets a ceiling per exposure that no photon count can beat. Stack short frames to go beyond it.

3.2 Extended sources (galaxies, nebulae, planets)

Set `Source geometry` to `extended` and give the source area in arcsec^2 (e.g. a $2' \times 1'$ galaxy ≈ 7200 ; Jupiter ≈ 1500 at opposition). Keep entering the *integrated* magnitude: the ETC spreads it uniformly over the stated area, so the aperture collects the corresponding area fraction and pixel saturation is judged on the surface brightness, which is what actually saturates on a resolved object. The approximation is intended for sources clearly larger than the seeing disc.

3.3 Target S/N mode

Enter a value in `Target S/N` to invert the calculation: the ETC solves for the exposure in closed form, including read noise, and the time-series plot switches to required exposure versus time. If the required exposure would saturate the brightest pixel, the result is flagged and the maximum unsaturated single-frame exposure is reported — split the integration into shorter frames and budget one read-noise term per frame.

4 Spectroscopy step by step

4.1 Slit spectroscopy

1. Mode `slit`; set the resolving power R (or load a measured slit-width/ R curve, which then overrides it), the slit width and the extraction height in arcsec , and pixels per resolution element.
2. Set the wavelength range and the S/N reference wavelength: the time-series evaluates that wavelength bin at every time sample.
3. Choose the **slit orientation**. `parallactic` assumes you rotate the slit to the parallactic angle (reported in the Results window for every time), avoiding atmospheric-dispersion losses. `fixed` applies the worst-case per-wavelength loss — expect tens of per cent missing at the blue end at airmass 1.5–2.
4. Run. The plot window shows the detected source rate and S/N per resolution element, with a manual time slider through the night.

4.2 Slitless spectroscopy and the Star Analyser

Mode `slitless`. Two ways to describe the disperser:

- **Star Analyser 100/200 (recommended)**: enter the groove density (100 or 200 lines/mm) and the grating-to-sensor distance in mm; the dispersion in $\text{\AA}/\text{pixel}$ follows from the physics (an SA100 at 42 mm on $4.63\ \mu\text{m}$ pixels gives $\simeq 11\ \text{\AA}/\text{pixel}$). Optionally set the first-order grating efficiency (~ 0.5 is realistic).
- **Manual**: leave the groove density at 0 and type the dispersion in $\text{\AA}/\text{pixel}$ directly.

The resolution element is set by seeing, the intrinsic grating LSF and atmospheric dispersion; with $2\text{--}3''$ seeing expect $R \sim 100\text{--}200$ regardless of exposure. The cross-dispersion extraction height is the only aperture; there are no slit losses. Slitless results are flagged *experimental* until validated on a calibrated instrument.

4.3 Reading the spectroscopic output

All spectroscopic counts and S/N are **per resolution element**, not per pixel. The Results table lists, per wavelength: the element width, the effective R , source and sky rates, S/N, peak electrons and ADU of the brightest pixel, the saturation flag and the maximum unsaturated exposure.

5 Results, plots and export

5.1 The Results window

Three tabs:

- **Selected-time result:** the numerical table for the chosen time (magnitude row in photometry; per-wavelength rows in spectroscopy).
- **Time series:** one row per time sample with UTC and local time, MJD, elevation, azimuth, **parallactic angle**, airmass (Pickering, refraction-corrected), S/N or required exposure, **sky surface brightness** actually used, band-effective extinction and its per-airmass coefficient, the absorption-adjusted apparent magnitude, saturation flag, peak electrons and maximum unsaturated exposure. This table is the CSV export.
- **Assumptions / visibility:** a plain-language statement of every physical assumption of the current run — sky model and its components, PSF model, noise terms, saturation policy — plus the selected-time altitude, airmass, parallactic angle and sky brightness.

5.2 Scientific plots

The native Matplotlib window provides pan/zoom/save, manual axis limits (time axes accept ISO timestamps), autoscale per panel, and in spectroscopy a manual slider that walks the sampled times of the observing interval, updating the spectrum panels and the markers. The sky-path panel shows target, Sun (daytime only) and Moon (only above the horizon) in altitude–azimuth (N=0°, E=90°).

5.3 Practical notes

- The Moon’s effect is included automatically in `ing` sky mode whenever it is above the horizon; within $\sim 30^\circ$ of a bright Moon expect the sky several magnitudes brighter.
- In `fixed_ab` mode the entered sky brightness is used as-is at every time — use it when you have a measured value.
- Airmass and all results are deliberately unavailable below 5° altitude.
- Save an **instrument profile** once your telescope/detector column is right; it restores automatically at start-up and travels with its QE/response files.

6 Troubleshooting

If a run aborts with a Python traceback, the full text is shown in the error dialog: it names the offending input explicitly and is the first thing to attach to a bug report, together with `etc_user_config.json`.

Table 1: Common messages and their resolution.

Message / symptom	Cause and fix
“...covers X–Y Å, but the requested calculation needs...”	A curve (template, QE, filter, atmosphere) does not cover the requested wavelengths. Narrow the range or load data with wider coverage; the ETC never extrapolates silently.
“Target altitude is below 5 deg”	The selected time puts the target too low. Move the time or check the coordinates; the visibility plot shows when the target rises.
“The template has no positive flux in the selected magnitude bandpass”	Reference filter outside the template’s spectral range (e.g. <i>K</i> band with an optical-only spectrum). Choose a filter inside the template coverage.
Required exposure flagged as saturating	The target S/N needs a longer exposure than the brightest pixel allows. Use the reported maximum unsaturated exposure and stack frames.
S/N of a bright star lower than expected	Scintillation ceiling (physical, not an error). Stack short exposures; larger aperture and lower airmass raise the ceiling.
“Resolve name” fails	SIMBAD needs internet; type coordinates manually — nothing else requires the network.
Filters missing at start-up	<code>data/filters.list</code> not found or empty. Each logical filter needs its <code>_Vega</code> and <code>_AB</code> XML profiles; <code>BLANK.xml</code> in <code>data/filters/</code> enables the unfiltered response.
Slit width outside the calibrated range	A measured slit- <i>R</i> curve is loaded and the requested width lies outside it. Change the width or remove the curve.